

Section-A

Some Basic Concepts of Chemistry

(Stoichiometry and Stoichiometric Calculations)

1. 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to
- Zero mg
 - 200 mg
 - 750 mg
 - 250 mg

(Mole Concept and Molar Masses)

2. The highest number of helium atoms is in
- 4 g of helium
 - 2.271098 L of helium at STP
 - 4 mol of helium
 - 4 u of helium

Structure of Atom

(Bohr's Model for Hydrogen Atom)

3. The energy of an electron in the ground state ($n = 1$) for He^+ ion is $-x\text{J}$, then that for an electron in $n = 2$ state for Be^{3+} ion in J is:
- $-4x$
 - $-\frac{4}{9}x$
 - $-x$
 - $-\frac{x}{9}$

(Quantum Mechanical Model of Atom)

4. Match List I with List II.

List I Quantum Number	List II Information provided
A. m_l	I. shape of orbital
B. m_s	II. size of orbital
C. l	III. orientation of orbital
D. n	IV. orientation of spin of electron

Choose the correct answer from the options given below:

- A-III, B-IV, C-II, D-I
- A-II, B-I, C-IV, D-III
- A-I, B-III, C-II, D-IV
- A-III, B-IV, C-I, D-II

Classification of Elements and Periodicity in Properties

(Periodic Trends in Properties of Elements)

5. Arrange the following elements in increasing order of electronegativity:

N, O, F, C, Si

Choose the correct answer from the options given below:

- $\text{O} < \text{F} < \text{N} < \text{C} < \text{Si}$
- $\text{F} < \text{O} < \text{N} < \text{C} < \text{Si}$
- $\text{Si} < \text{C} < \text{N} < \text{O} < \text{F}$
- $\text{Si} < \text{C} < \text{O} < \text{N} < \text{F}$

6. Arrange the following elements in increasing order of first ionization enthalpy:

Li, Be, B, C, N

Choose the correct answer from the options given below:

- $\text{Li} < \text{Be} < \text{C} < \text{B} < \text{N}$
- $\text{Li} < \text{Be} < \text{N} < \text{B} < \text{C}$
- $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{N}$
- $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N}$

Chemical Bonding and Molecular Structure

(VSEPR Theory)

7. Match List I with List II.

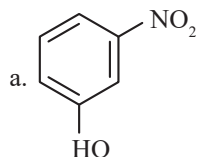
List I (Compound)	List II (Shape/geometry)
A. NH_3	I. Trigonal Pyramidal
B. BrF_5	II. Square Planar
C. XeF_4	III. Octahedral
D. SF_6	IV. Square Pyramidal

Choose the correct answer from the options given below:

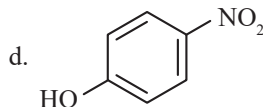
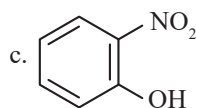
- A-III, B-IV, C-I, D-II
- A-II, B-III, C-IV, D-I
- A-I, B-IV, C-II, D-III
- A-II, B-IV, C-III, D-I

(Hydrogen Bonds)

8. Intramolecular hydrogen bonding is present in



b. HF

**Thermodynamics**

(Thermodynamics Terms)

9. Match List I with List II.

List I (Process)		List II (Conditions)	
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:

- a. A-I, B-II, C-III, D-IV b. A-II, B-III, C-IV, D-I
c. A-IV, B-III, C-II, D-I d. A-IV, B-II, C-III, D-I

(Spontaneity, Entropy, Gibbs Energy Change and Equilibrium)

10. In which of the following processes entropy increases?

- A. A liquid evaporates to vapour
B. Temperature of a crystalline solid lowered from 130K to 0 K.
C. $2 \text{NaHCO}_{3(s)} \rightarrow \text{Na}_2\text{CO}_{3(s)} + \text{CO}_{2(g)} + \text{H}_2\text{O}_{(g)}$
D. $\text{Cl}_{2(g)} \rightarrow 2 \text{Cl}_{(g)}$

Choose the correct answer from the options given below:

- a. A, C and D b. C and D
c. A and C d. A, B and D

Equilibrium

(Law of Chemical Equilibrium, Equilibrium Const. and Its Application)

11. For the reaction $2\text{A} \rightleftharpoons \text{B} + \text{C}$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is:

$$[\text{A}] = [\text{B}] = [\text{C}] = 2 \times 10^{-3} \text{ M.}$$

Then, which of the following is correct?

- a. Reaction has a tendency to go in backward direction.
b. Reaction has gone to completion in forward direction.
c. Reaction is at equilibrium.
d. Reaction has a tendency to go in forward direction.

12. In which of the following equilibria, K_p and K_c are NOT equal?

- a. $\text{CO}_{(g)} + \text{H}_2\text{O}_{(g)} \rightleftharpoons \text{CO}_{2(g)} + \text{H}_{2(g)}$
b. $2\text{BrCl}_{(g)} \rightleftharpoons \text{Br}_{2(g)} + \text{Cl}_{2(g)}$
c. $\text{PCl}_{5(g)} \rightleftharpoons \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$
d. $\text{H}_{2(g)} + \text{I}_{2(g)} \rightleftharpoons 2 \text{HI}_{(g)}$

Redox Reactions

(Redox Reactions (Oxidation and Reduction), Oxidation Number)

13. Which reaction is NOT a redox reaction?

- a. $\text{H}_2 + \text{Cl}_2 \rightarrow 2 \text{HCl}$
b. $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2 \text{NaCl}$
c. $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
d. $2 \text{KClO}_3 + \text{I}_2 \rightarrow 2 \text{KIO}_3 + \text{Cl}_2$

Organic Chemistry-Some Basic Principles and Techniques

(Classification, Nomenclature of Organic Compounds)

14. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is:

- a. 2,3-dimethylbutane b. 2,2-dimethylbutane
c. n-hexane d. 2-methylpentane

(Tetravalence of Carbon (Shapes of Organic Compounds))

15. Match List I with List II.

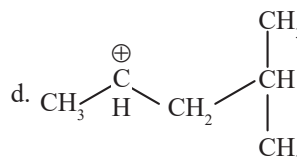
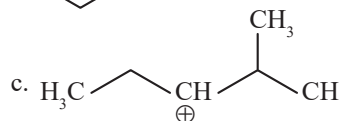
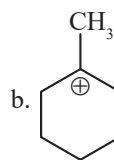
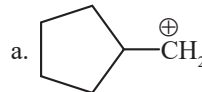
List I (Molecule)		List II (Number and types of bonds between two carbon atoms)	
A.	ethane	I.	one σ -bond and two π -bonds
B.	ethene	II.	two π -bonds
C.	carbon molecule, C_2	III.	one σ -bond
D.	ethyne	IV.	one σ -bond and one π -bond

Choose the correct answer from the options given below:

- a. A-III, B-IV, C-II, D-I b. A-III, B-IV, C-I, D-II
c. A-I, B-IV, C-II, D-III d. A-IV, B-III, C-II, D-I

(Fundamental Concepts of Organic Reactions)

16. The most stable carbocation among the following is:



(Methods of Purification, Qualitative and Quantitative Analysis of Organic Compounds)

17. On heating some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as

a. Distillation b. Chromatography
c. Crystallization d. Sublimation

Hydrocarbons

(Alkanes)

18. Given below are two statements:

Statement I: The boiling point of three isomeric pentanes follows the order

n-pentane > isopentane > neopentane

Statement II: When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- a. Statement I is correct but Statement II is incorrect.
b. Statement I is incorrect but Statement II is correct.
c. Both Statement I and Statement II are correct.
d. Both Statement I and Statement II are incorrect.

Solutions

(Solubility)

19. The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order:

- a. $A > C > B$ b. $A > B > C$
c. $B > A > C$ d. $B > C > A$

Electrochemistry

(Electrolytic Cells and Electrolysis)

20. Match List I with List II.

List I (Conversion)		List II (Number of Faraday required)	
A.	1 mol of H_2O to O_2	I.	3F
B.	1 mol of MnO_4^- to Mn^{2+}	II.	2F
C.	1.5 mol of Ca from molten $CaCl_2$	III.	1F
D.	1 mol of FeO to Fe_2O_3	IV.	5F

Choose the correct answer from the options given below:

- a. A-II, B-III, C-I, D-IV b. A-III, B-IV, C-II, D-I
c. A-II, B-IV, C-I, D-III d. A-III, B-IV, C-I, D-II

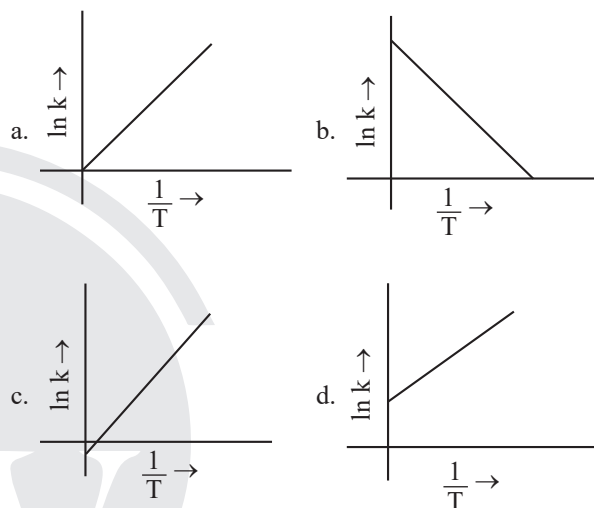
Chemical Kinetics

(Temperature Dependence of the Rate of a Reaction & Effect of Catalysts)

21. Activation energy of any chemical reaction can be calculated if one knows the value of

- a. orientation of reactant molecules during collision
b. rate constant at two different temperatures
c. rate constant at standard temperature
d. probability of collision

22. Which plot of $\ln k$ vs $\frac{1}{T}$ is consistent with Arrhenius equation?



The p-Block Elements (Group 15 to 18)

(Group 16 Elements (Oxygen Family))

23. Given below are two statements:

Statement I: The boiling point of hydrides of Group 16 elements follow the order

$H_2O > H_2Te > H_2Se > H_2S$.

Statement II: On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O , it has higher boiling point.

In the light of the above statements, choose the *correct* answer from the options given below:

- a. Statement I is true but Statement II is false.
b. Statement I is false but Statement II is true.
c. Both Statement I and Statement II are true.
d. Both Statement I and Statement II are false.

24. Among Group 16 elements, which one does **NOT** show -2 oxidation state?

- a. Te b. Po
c. O d. Se

The d- and f-Block Elements

(Transition Elements (d-Block))

25. The E° value for the $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is more positive than that of $\text{Cr}^{3+}/\text{Cr}^{2+}$ or $\text{Fe}^{3+}/\text{Fe}^{2+}$ due to change of
- d^4 to d^5 configuration
 - d^3 to d^5 configuration
 - d^5 to d^4 configuration
 - d^5 to d^2 configuration
26. 'Spin only' magnetic moment is same for which of the following ions?
- Ti^{3+}
 - Cr^{2+}
 - Mn^{2+}
 - Fe^{2+}
 - Sc^{3+}

Choose the most appropriate answer from the options given below:

- B and C only
- A and D only
- B and D only
- A and E only

Coordination Compounds

(Nomenclature & Isomerism in Coordination Compounds)

27. Match List-I with List-II.

List-I (Complex)		List-II (Type of isomerism)
A. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	I.	Solvate isomerism
B. $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$	II.	Linkage isomerism
C. $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$	III.	Ionization isomerism
D. $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$	IV.	Coordination isomerism

Choose the correct answer from the options given below:

- A-I, B-IV, C-III, D-II
- A-II, B-IV, C-III, D-I
- A-II, B-III, C-IV, D-I
- A-I, B-III, C-IV, D-II

(Bonding in Coordination Compounds)

28. Given below are two statements:

Statement I: Both $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.

Statement II: $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic whereas $[\text{CoF}_6]^{3-}$ is paramagnetic.

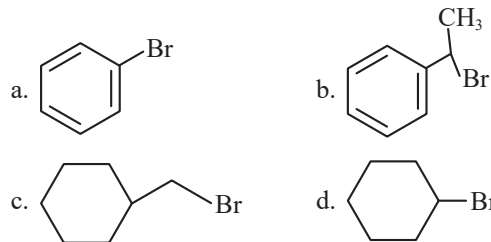
In the light of the above statements, choose the correct answer from the options given below:

- Statement I is true but Statement II is false.
- Statement I is false but Statement II is true.
- Both Statement I and Statement II are true.
- Both Statement I and Statement II are false.

Haloalkanes and Haloarenes

(Haloarenes and Chemical Reactions of Haloarenes)

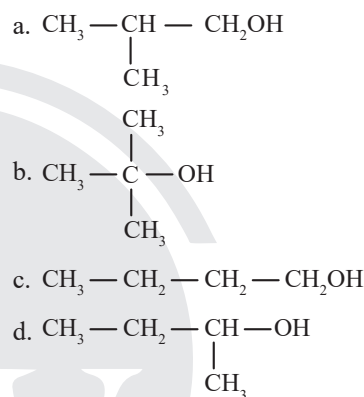
29. The compound that will undergo $\text{S}_{\text{N}}1$ reaction with the fastest rate is



Alcohols, Phenols and Ethers

(Alcohols)

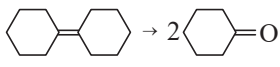
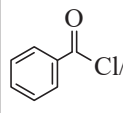
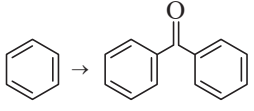
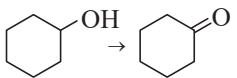
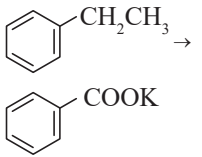
30. Which one of the following alcohols reacts instantaneously with Lucas reagent?



Aldehydes, Ketones and Carboxylic Acids

(Preparation of Aldehydes and Ketones)

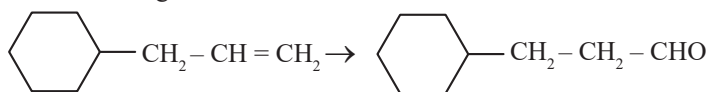
31. Match List I with List II.

List I (Reaction)	List II (Reagents/Condition)
A. 	I.  Anhyd. AlCl_3
B. 	II. CrO_3
C. 	III. $\text{KMnO}_4/\text{KOH}, \Delta$
D. 	IV. (i) O_3 (ii) $\text{Zn-H}_2\text{O}$

Choose the correct answer from the options given below:

- A-IV, B-I, C-II, D-III
- A-I, B-IV, C-II, D-III
- A-IV, B-I, C-III, D-II
- A-III, B-I, C-II, D-IV

32. Identify the correct reagents that would bring about the following transformation.



- a. (i) BH₃
 (ii) H₂O₂/OH[⊖]
 (iii) alk. KMnO₄
 (iv) H₃O[⊕]
 b. (i) H₂O/H⁺
 (ii) PCC
 c. (i) H₂O/H⁺
 (ii) CrO₃
 d. (i) BH₃
 (ii) H₂O₂/OH[⊖]
 (iii) PCC

(Physical Properties and Chemical Reactions of Aldehydes and Ketones)

33. Fehling's solution 'A' is

- a. alkaline solution of sodium potassium tartrate (Rochelle's salt)
 b. aqueous sodium citrate
 c. aqueous copper sulphate
 d. alkaline copper sulphate

Amines

(Physical & Chemical Properties of Amines)

34. Given below are two statements:

Statement I: Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II: Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement I is correct but Statement II is false.
 b. Statement I is incorrect but Statement II is true.
 c. Both Statement I and Statement II are true.
 d. Both Statement I and Statement II are false.

Biomolecules

(Carbohydrates)

35. The reagents with which glucose does **not** react to give the corresponding tests/products are

- A. Tollen's reagent B. Schiff's reagent
 C. HCN D. NH₂OH
 E. NaHSO₃

Choose the correct options from the given below:

- a. B and E b. E and D
 c. B and C d. A and D

Section-B

Some Basic Concepts of Chemistry

(Percentage Composition, Empirical & Molecular Formula)

36. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is:
 (Given atomic masses of A = 64; B = 40; C = 32u)
- a. AB₂C₂ b. ABC₄
 c. A₂BC₂ d. ABC₃

Chemical Bonding and Molecular Structure

(Bond Parameters and Dipole moment)

37. Identify the correct answer.
- a. Dipole moment of NF₃ is greater than that of NH₃.
 b. Three canonical forms can be drawn for CO₃²⁻ ion
 c. Three resonance structures can be drawn for ozone.
 d. BF₃ has non-zero dipole moment.

Thermodynamics

(Measurement of ΔU & ΔH, Work Done and Heat Capacity)

38. The work done during reversible isothermal expansion of one mole of hydrogen gas at 25 °C from pressure of 20 atmosphere to 10 atmosphere is:
 (Given R = 2.0 cal K⁻¹ mol⁻¹)
- a. 413.14 calories b. 100 calories
 c. 0 calorie d. - 413.14 calories

Equilibrium

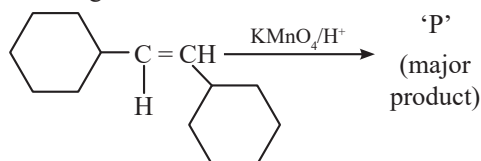
(Ionization of Acids & Bases, pH Scale, Hydrolysis)

39. Consider the following reaction in a sealed vessel at equilibrium with concentrations of N₂ = 3.0 × 10⁻³ M, O₂ = 4.2 × 10⁻³ M and NO = 2.8 × 10⁻³ M .
- $$2\text{NO}_{(g)} \rightleftharpoons \text{N}_{2(g)} + \text{O}_{2(g)}$$
- If 0.1 mol L⁻¹ of NO_(g) is taken in a closed vessel, what will be degree of dissociation (α) of NO_(g) at equilibrium ?
- a. 0.8889 b. 0.717
 c. 0.00889 d. 0.0889

Hydrocarbons

(Alkenes)

40. For the given reaction



'P' is

- a.
- b.
- c.
- d.

Solutions

(Colligative Properties and Determination of Molar Mass)

41. The plot of osmotic pressure (Π) vs concentration (mol L^{-1}) for a solution gives a straight line with slope $25.73 \text{ L bar mol}^{-1}$. The temperature at which the osmotic pressure measurement is done is:

(Use $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$)

- a. 25.73°C b. 12.05°C
c. 37°C d. 310°C

Electrochemistry

(Electrolytic Cells and Electrolysis)

42. Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is:

(Given : Molar mass of Cu : 63 g mol^{-1} , $1 \text{ F} = 96487 \text{ C}$)

- a. 31.5 g b. 0.0315 g
c. 3.15 g d. 0.315 g

Chemical Kinetics

(Temperature Dependence of the Rate of a Reaction & Effect of Catalysts)

43. The rate of a reaction quadruples when temperature changes from 27°C to 57°C . Calculate the energy of activation.

Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $\log 4 = 0.6021$

- a. 3.80 kJ/mol b. 3804 kJ/mol
c. 38.04 kJ/mol d. 380.4 kJ/mol

The d- and f-Block Elements

(Inner Transition Elements (Lanthanoids and Actinoids))

44. The pair of lanthanoid ions which are diamagnetic is

- a. Gd^{3+} and Eu^{3+} b. Pm^{3+} and Sm^{3+}
c. Ce^{4+} and Yb^{2+} d. Ce^{3+} and Eu^{2+}

Coordination Compounds

(Introduction and Werner's Theory of Coordination Compounds)

45. Given below are two statements:

Statement I: $[\text{Co}(\text{NH}_3)_6]^{3+}$ is a homoleptic complex whereas $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is a heteroleptic complex.

Statement II: Complex $[\text{Co}(\text{NH}_3)_6]^{3+}$ has only one kind of ligands but $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ has more than one kind of ligands.

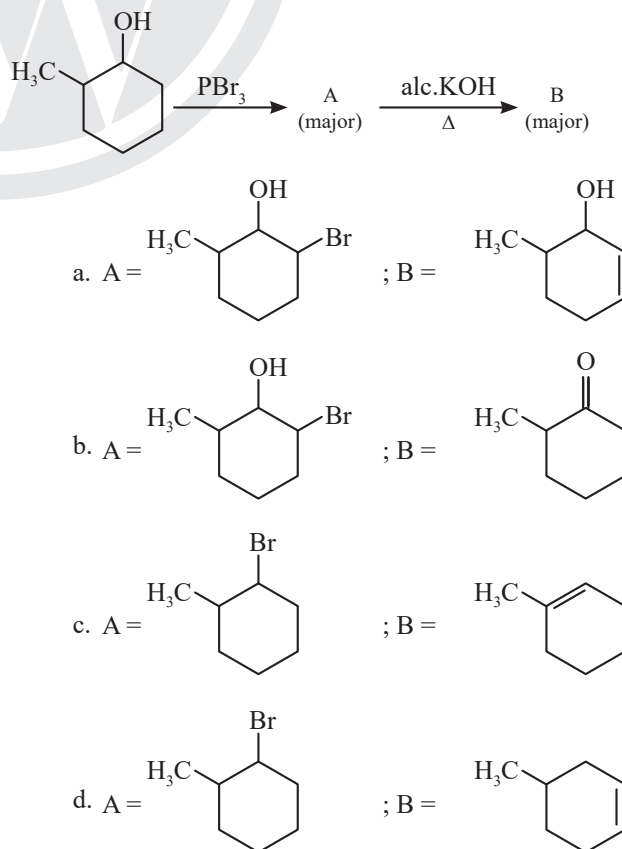
In the light of the above statements, choose the correct answer from the options given below:

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b. Statement I is false but Statement II is true.
c. Both Statement I and Statement II are true.
d. Both Statement I and Statement II are false.

Haloalkanes and Haloarenes

(Haloalkanes and Chemical Reactions of Haloalkanes)

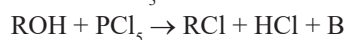
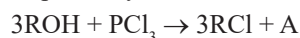
46. Major products A and B formed in the following reaction sequence, are



Alcohols, Phenols and Ethers

(Alcohols)

47. The products A and B obtained in the following reactions, respectively, are

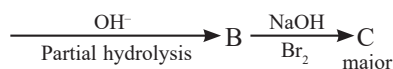
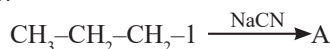


- a. H_3PO_4 and POCl_3 b. H_3PO_3 and POCl_3
 c. POCl_3 and H_3PO_3 d. POCl_3 and H_3PO_4

Amines

(Preparation of Amines)

48. Identify the major product C formed in the following reaction sequence:



- a. butanamide b. α -bromobutanoic acid
 c. propylamine d. butylamine

Principle Related to Practical Chemistry

(Preparation of Organic and Inorganic Compounds)

49. During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

- a. dilute nitric acid
 b. dilute sulphuric acid
 c. dilute hydrochloric acid
 d. concentrated sulphuric acid

(Qualitative Analysis of Inorganic Salts)

50. Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.

- A. Al^{3+} B. Cu^{2+}
 C. Ba^{2+} D. Co^{2+}
 E. Mg^{2+}

Choose the correct answer from the options given below:

- a. E, C, D, B, A b. E, A, B, C, D
 c. B, A, D, C, E d. B, C, A, D, E

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (c) | 3. (c) | 4. (d) | 5. (c) | 6. (d) | 7. (c) | 8. (c) | 9. (b) | 10. (a) |
| 11. (a) | 12. (c) | 13. (b) | 14. (a) | 15. (a) | 16. (b) | 17. (d) | 18. (c) | 19. (d) | 20. (c) |
| 21. (b) | 22. (b) | 23. (c) | 24. (b) | 25. (c) | 26. (c) | 27. (c) | 28. (c) | 29. (b) | 30. (b) |
| 31. (a) | 32. (d) | 33. (c) | 34. (c) | 35. (a) | 36. (d) | 37. (b) | 38. (d) | 39. (b) | 40. (d) |
| 41. (c) | 42. (d) | 43. (c) | 44. (c) | 45. (c) | 46. (c) | 47. (b) | 48. (c) | 49. (b) | 50. (c) |

Explanations

1. (d) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

Mass of NaOH = 1 g

$$\text{Moles of NaOH} = \frac{1}{40} \text{ moles}$$

= 0.025 moles

= 25 millimoles

$$\text{Millimoles of HCl} = M \times V = 25 \times 0.75$$

= 18.75 millimoles

Millimoles of NaOH unreacted

$$= 25 - 18.75$$

= 6.25 millimoles

$$\text{Mass} = 40 \times 6.25 = 250 \text{ mg}$$

2. (c)

$$\text{Atoms in 4 g of Helium} = \frac{4}{4} \times N_A = N_A$$

$$\text{Atoms in 2.271098 L of Helium at STP} = \frac{2.271098}{22.4} \times N_A$$

$$= 0.101 N_A$$

$$\text{Atoms in 4 mol of Helium} = 4 \times N_A = 4N_A$$

$$\text{Atoms in 4u of Helium} = \frac{4}{4} = 1$$

Hence, highest number of atoms are present in 4 mol of Helium.

3. (c)

$$\text{For He}^+ \text{ ion; Energy} = -x \text{ J, } Z = 2, n = 1$$

$$\text{Energy} = \frac{-2.18 \times 10^{-18} Z^2}{n^2} \text{ J}$$

$$-x = \frac{-2.18 \times 10^{-18} \times (2)^2}{(1)^2} \text{ J}$$

$$x = 2.18 \times 10^{-18} \times 4 \text{ J}$$

- For Be^{3+} ion; $Z = 4, n = 2$

$$(E_2)_{\text{Be}^{3+}} = \frac{-2.18 \times 10^{-18} Z^2}{n^2} \text{ J}$$

$$= \frac{-2.18 \times 10^{-18} \times (4)^2}{(2)^2} \text{ J}$$

$$= -2.18 \times 10^{-18} \times 4 \text{ J}$$

$$= -x \text{ J}$$

4. (d)

- (A) $m_l \rightarrow$ orientation of orbital.
- (B) $m_s \rightarrow$ orientation of spin of electron.
- (C) $l \rightarrow$ shape of orbital.
- (D) $n \rightarrow$ size of orbital.

5. (c)

$$\left. \begin{array}{l} \text{F} = 4.0 \\ \text{O} = 3.5 \\ \text{N} = 3.0 \\ \text{C} = 2.5 \\ \text{Si} = 1.8 \end{array} \right\} \text{electronegativity}$$

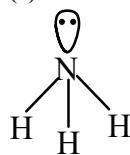
Hence order will be:

$$\text{Si} < \text{C} < \text{N} < \text{O} < \text{F}$$

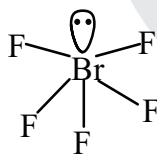
6. (d)

- In a period, generally on moving left to right ionization enthalpy increases. But due to more stable configuration of $\text{Be}(2s^2)$, its ionization enthalpy is more as compared to $\text{B}(2p^1)$, less stable configuration.
- So, correct order of first IE : $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{N}$

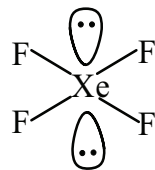
7. (c)



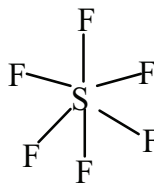
Trigonal
Pyramidal



Square
Pyramidal

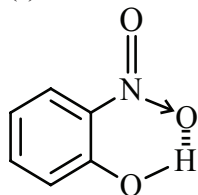


Square
Planar



Octahedral

8. (c)



Intramolecular hydrogen bonding is observed in ortho nitrophenol.

9. (b) Isothermal process $\rightarrow$ at constant temperature

Isochoric process $\rightarrow$ at constant volume

Isobaric process $\rightarrow$ at constant pressure

Adiabatic process $\rightarrow$ No heat exchange

10. (a)

- (A) When liquid evaporates to vapour, entropy increases due to increase in randomness.
- (B) When temperature of crystalline solid lowered from 130 K to 0 K, the random collisions will decrease, resulting in decrease in entropy.
- (C) In the given reaction,

$$2\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$$
 Solid is converting to gas, due to which randomness increases and hence, entropy increase.
- (D) In the given conversion, $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$
 number of gaseous atom increases, due to which entropy increases.

Hence, Entropy will increase in case of A, C, D.

11. (a)



$$Q_c = \frac{[\text{B}][\text{C}]}{[\text{A}]^2} = \frac{[2 \times 10^{-3}][2 \times 10^{-3}]}{[2 \times 10^{-3}]^2}$$

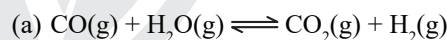
$$Q_c = 1$$

$$Q_c > K_c$$

Reaction has a tendency to go in backward direction.

$Q_c < K_c$ Reaction has a tendency to go in forward direction.

12. (c)



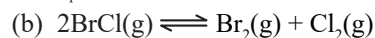
$$K_c = \frac{[\text{CO}_2][\text{H}_2]}{[\text{CO}][\text{H}_2\text{O}]}$$

$$\Delta n_g = 2 - 2 = 0$$

$$K_p = K_c (\text{RT})^{\Delta n_g}$$

$$K_p = K_c (\text{RT})^0$$

$$K_p = K_c$$



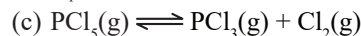
$$\Delta n_g = 2 - 2 = 0$$

$$K_c = \frac{[\text{Br}_2][\text{Cl}_2]}{[\text{BrCl}]^2}$$

$$K_p = K_c (\text{RT})^{\Delta n_g}$$

$$K_p = K_c (\text{RT})^0$$

$$K_p = K_c$$



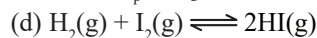
$$\Delta n_g = 2 - 1 = 1$$

$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]}$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$K_p = K_c (RT)^1$$

$$\text{Here, } K_p \neq K_c$$



$$\Delta n_g = 2 - 2 = 0$$

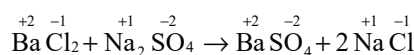
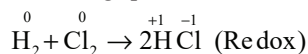
$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

$$K_p = K_c (RT)^{\Delta n_g}$$

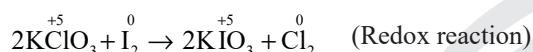
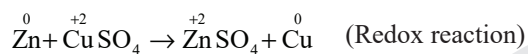
$$K_p = K_c (RT)^0$$

$$\text{Here, } K_p = K_c$$

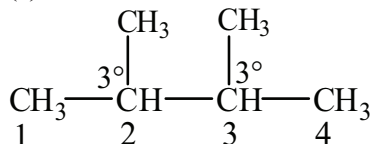
13. (b) Redox reactions involve change in oxidation number of the interacting species.



(No change in oxidation state, hence not redox reaction)

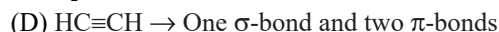
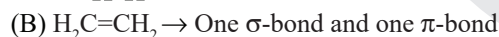
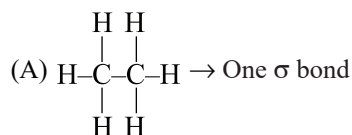


14. (a)

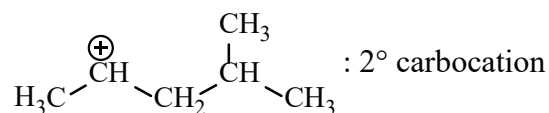
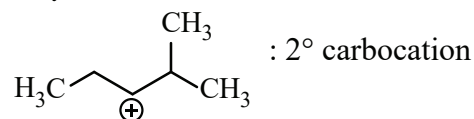
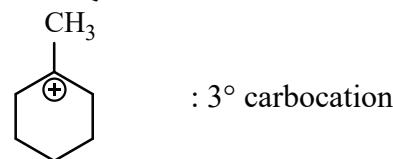
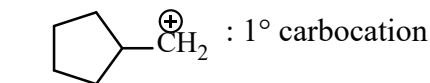


IUPAC name: 2,3-dimethyl butane

15. (a)



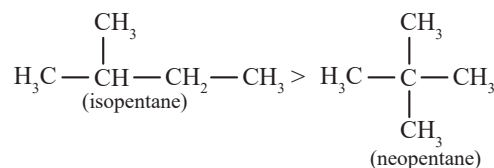
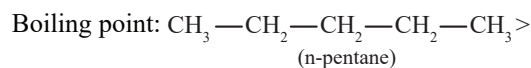
16. (b)



- Stability order of carbocation : $3^\circ > 2^\circ > 1^\circ$.
- Hence, most stable carbocation is present in option (b).

17. (d) On heating, solid substances change from solid to vapour state without passing through liquid state. This purification technique is known as sublimation and is used to separate sublimable compound from non-sublimable impurities.

18. (c) Boiling point decreases with increase in branching as the intermolecular forces decreases

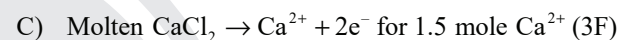
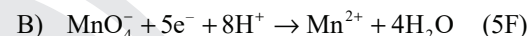
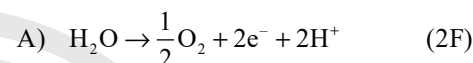


19. (d) Lesser the value of K_H , more will be the solubility of gas in water.

So, solubility order:

$$\text{B}(K_H = 2 \times 10^{-5}) > \text{C}(K_H = 35) > \text{A}(K_H = 145)$$

20. (c)



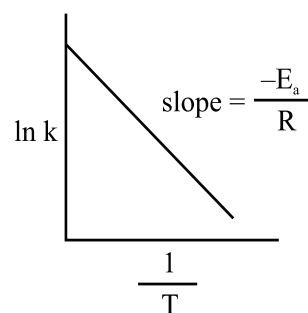
21. (b) Activation energy of any chemical reaction can be calculated if one knows the value of rate constant at two different temperature by given formula:

$$\log \frac{k_1}{k_2} = \frac{E_a}{2.303R} \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$$

22. (b) Arrhenius equation is;

$$\ln k = -\frac{E_a}{R} \times \frac{1}{T} + \log A$$

Graph between $\ln k$ vs $\frac{1}{T}$ with intercept = $\log A$ and slope = $-\frac{E_a}{R}$



23. (c) The boiling point of hydrides of group 16 are in the order $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$.

It is due to the presence of extensive H-bonding in H_2O .

24. (b) Po does not show -2 oxidation state because on going down the group, electropositive nature increases.

25. (c) Much larger third ionisation energy of Mn (where the required change is d^5 to d^4) is mainly responsible for this. This also explains why the $+3$ state of Mn is of little importance.

26. (c) Spin only magnetic moment

$$= \sqrt{n(n+2)}$$

n = number of unpaired electrons

A	Ti ³⁺	d ¹	1.73
B	Cr ²⁺	d ⁴	4.90
C	Mn ²⁺	d ⁵	5.92
D	Fe ²⁺	d ⁶	4.90
E	Sc ³⁺	d ⁰	0

B = Cr²⁺ = 4.90 D = Fe²⁺ = 4.90

27. (c)

(A) [Co(NH₃)₅(NO₂)]Cl₂ shows linkage isomerism. Its linkage isomer is [Co(NH₃)₅(ONO)]Cl₂(B) [Co(NH₃)₅(SO₄)]Br shows ionization isomerism. Its ionization isomer is [Co(NH₃)₅(Br)]SO₄(C) [Co(NH₃)₆][Cr(CN)₆] shows coordination isomerism. Its coordination isomer is [Cr(NH₃)₆][Co(CN)₆](D) [Co(H₂O)₆]Cl₃ shows solvate isomerism. Its one of the solvate isomer is [Co(H₂O)₅Cl]Cl₂·H₂O

So correct option is:

A → II, B → III, C → IV, D → I

28. (c)

Both [Co(NH₃)₆]³⁺ and [CoF₆]³⁻ are octahedral complex because coordination number are six.Electronic configuration of Co³⁺ in [Co(NH₃)₆]³⁺ isCo³⁺: [Ar]3d⁶4s⁰4p⁰

3d	4s	4p
$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$		

As NH₃ is a strong ligand, so pairing takes place

3d	4s	4p
$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$ $\uparrow$		
n = 0		

d²sp³ hybridisation

No unpaired electron, so it is diamagnetic in nature

Electronic configuration of Co³⁺ in [CoF₆]³⁻ isCo³⁺: [Ar]3d⁶4s⁰4p⁰

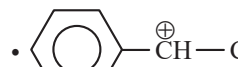
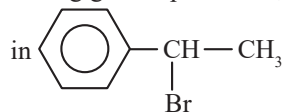
3d	4s	4p
$\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$		

n = 4

As F⁻ is a weak ligand, therefore no pairing takes place

So, it is paramagnetic in nature.

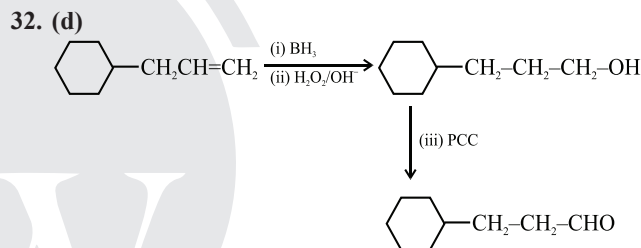
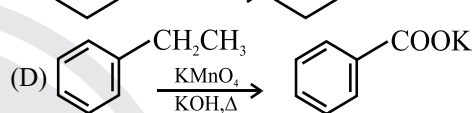
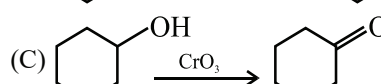
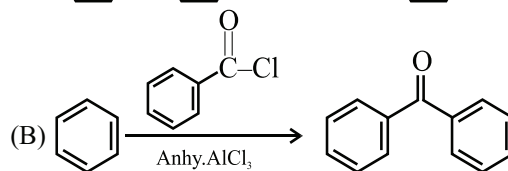
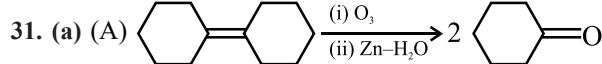
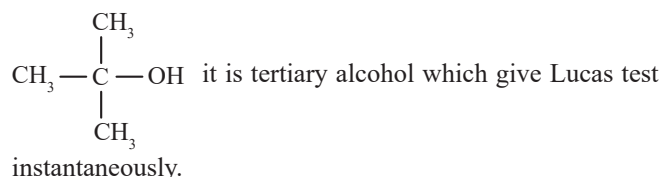
29. (b)

• S_N1 reaction are fastest in those compound, where the intermediate carbocation is resonance stabilised.•  is resonance stabilised carbocationamong given options. So, rate of S_N1 reaction will be fastest

30. (b) 3° alcohols react with Lucas reagent instantaneously.

2° alcohols react with Lucas reagent after 5 minute.

1° alcohols do not react with Lucas reagent at ordinary temperature.



33. (c) Fehling's solution 'A' is aqueous copper sulphate.

34. (c)

• Aniline does not undergo Friedel-Craft reaction due to salt formation with anhy. aluminium chloride (the lewis acid) which is used as a catalyst.

• Gabriel synthesis is used for the preparation of aliphatic primary amines only. Aromatic primary amines cannot be prepared by this method because aryl halide do not undergo nucleophilic substitution reactions.

35. (a) Despite having the aldehyde group, glucose does not give Schiff's test and it does not form the hydrosulphite addition product with NaHSO₃.

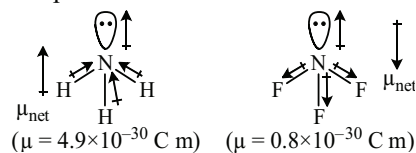
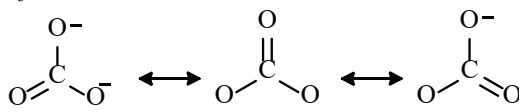
36. (d)

Element	Percentage composition	Atomic mass	Atomic ratio of atoms	Simplest ratio of atoms
A	32%	64	$\frac{32}{64} = 0.5$	$\frac{0.5}{0.5} = 1$
B	20%	40	$\frac{20}{40} = 0.5$	$\frac{0.5}{0.5} = 1$
C	48%	32	$\frac{48}{32} = 1.5$	$\frac{1.5}{0.5} = 3$

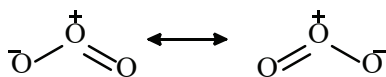
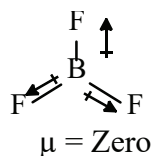
Hence, empirical formula is ABC₃.

37. (b)

• Dipole Moment:

• CO_3^{2-} has three canonical forms.

• Two resonating structures can be drawn for ozone.

• BF_3 has trigonal planar structures. Its net dipole moment is Zero.

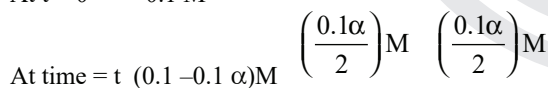
38. (d) $W = -2.303nRT \log \left(\frac{P_1}{P_2} \right)$

$$= -2.303 \times 1 \times 2 \times 298 \times \log \left(\frac{20}{10} \right)$$

$$= -2.303 \times 2 \times 298 \times 0.301$$

$$= -413.14 \text{ Calories}$$

39. (b)

At $t = 0$ 0.1 M

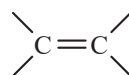
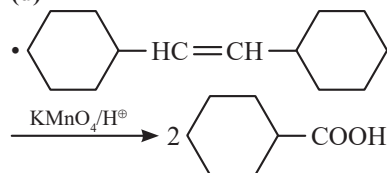
$$\therefore K_c = \frac{3 \times 10^{-3} \times 4.2 \times 10^{-3}}{(2.8 \times 10^{-3})^2} = \frac{0.1\alpha \times 0.1\alpha}{(0.1 - 0.1\alpha)^2}$$

$$1.6 = \frac{\left(\frac{0.1\alpha}{2} \right)^2}{(0.1 - 0.1\alpha)^2}$$

$$1.27 = \frac{\left(\frac{0.1\alpha}{2} \right)}{0.1 - 0.1\alpha}$$

$$\alpha = 0.717$$

40. (d)



Will be completely oxidized in presence of acidified KMnO_4 into carboxylic acid.

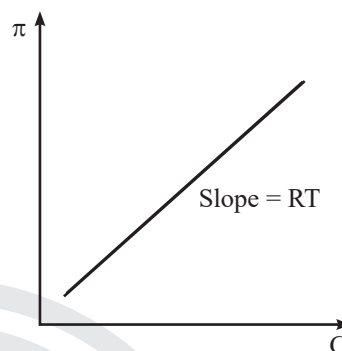
41. (c) $\pi = CRT$

$$\therefore \text{slope} = RT = 25.73 \text{ L bar mol}^{-1}$$

$$\therefore T = \frac{25.73}{0.083}$$

$$= 310\text{K}$$

$$= 37^\circ\text{C}$$



42. (d)

$$W = Z \times i \times t$$

$$= \frac{63}{96487 \times 2} \times 9.6487 \times 100$$

$$= 0.315\text{g}$$

43. (c)

$$\log \left(\frac{k_{330}}{k_{300}} \right) = \frac{E_a}{2.303R} \times \left[\frac{330 - 300}{300 \times 330} \right]$$

$$\text{Given: } \frac{k_{330}}{k_{300}} = 4$$

$$\therefore \log 4 = \frac{E_a}{2.303R} \times \frac{30}{300 \times 330}$$

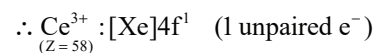
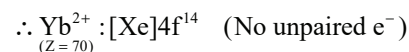
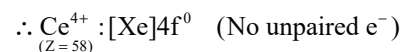
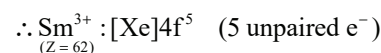
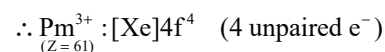
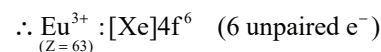
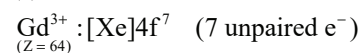
$$\therefore E_a = \frac{0.602 \times 2.303 \times 8.314 \times 300 \times 330}{30} \text{ J/mol}$$

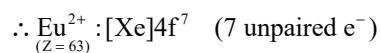
$$= 38037.7 \text{ J/mol}$$

$$= 38.0377 \text{ kJ/mol}$$

$$\approx 38.04 \text{ kJ/mol}$$

44. (c)





Species having no unpaired e^- [Ce^{4+} and Yb^{2+}] are diamagnetic in nature.

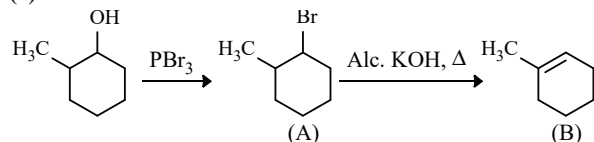
45. (c)

- Homoleptic complex is one which contains only one kind of ligands. Heteroleptic complex is one which contains two or more kinds of ligand.

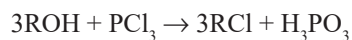
$[\text{Co}(\text{NH}_3)_6]^{3+}$ has only one kind of ligands but

$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ has more than one kind of ligands.

46. (c)



47. (b)



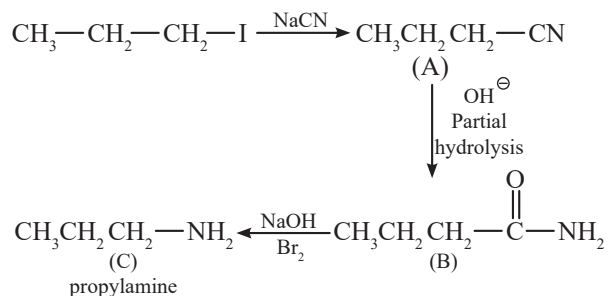
(A)



(B)

Hence, A : H_3PO_3 and B : POCl_3

48. (c)



49. (b) Fe^{2+} ion undergoes hydrolysis, therefore, while preparing aqueous solution of ferrous sulphate in water, 2-3 ml of dilute sulphuric acid is added to prevent the hydrolysis of Mohr's salt.

50. (c) Below given the ions and the group they belong as per qualitative analysis.

Cation	Group
Cu^{2+}	2
Al^{3+}	3
Co^{2+}	4
Ba^{2+}	5
Mg^{2+}	6